

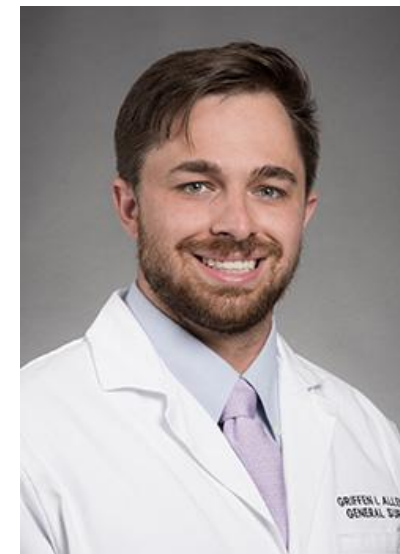
SONOMA Study

Objectives and Methods

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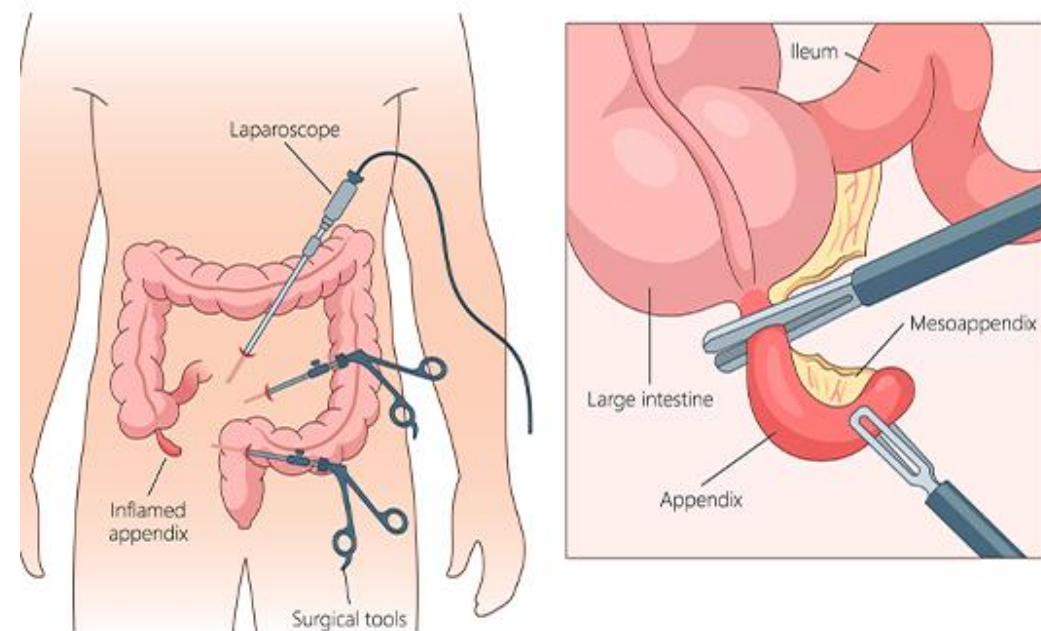
W DEPARTMENT OF BIostatISTICS
UNIVERSITY *of* WASHINGTON



Background

- The appendix is a small tube connected to the large intestine
- When this organ becomes inflamed or infected it's common to perform an urgent surgery called an appendectomy
- If untreated, the appendix could burst leading to potential life-threatening symptoms

Laparoscopic Appendectomy



Terminology

NOM: Non-operative management (managed with antibiotics)

Appendicolith: calcified deposit made of packed stool and mineral deposits that forms within the appendix.

Phlegmon: contained, diffuse inflammatory mass that forms around an inflamed appendix

Abscess: a collection of pus that forms as a complication of acute appendicitis or a ruptured appendix

NLR: (Neutrophil to lymphocyte ratio) a marker of systemic inflammation

Complicated Appendicitis: An appendicitis with an abscess, phlegmon or other complications



Comparison of Outcome of Antibiotic Drugs and Appendectomy (CODA) Study

A noninferiority randomized trial comparing 10-day antibiotic treatment with appendectomy in patients with appendicitis at 25 U.S centers.

- 1552 patients were evenly randomized
 - Antibiotics were noninferior to appendectomy on the basis of 30-day Euro Quality of Life Questionnaire scores
 - In the antibiotics group, 29% underwent an appendectomy by 90 days after NOM
 - Participants with an appendicolith were at a higher risk for appendectomy and for complications than those without an appendicolith
-

CODA
Study



National guidelines are
updated to include antibiotics
as an initial treatment for
appendicitis



TRIAD
Project

Treatment-Individualized Appendicitis Decision Making (TRIAD)

A project working to establish a decision aid for adults with appendicitis and training for clinicians to support informed decision making about the use of antibiotics and surgery for appendicitis

- 18 hospitals in 11 states
- Training ER staff to offer AppyOrNot tool and integrating it into standard practice

The screenshot shows the TRIAD decision aid interface. At the top, the TRIAD logo is on the left, and navigation links for 'English', a volume icon, and a settings icon are on the right. The page is labeled 'Page 2 of'. The main content area has a light blue header bar. Below it, the first question is 'Are you a:' with a red asterisk and 'must provide value'. To the right are four buttons: 'Patient diagnosed with appendicitis', 'Family member or loved one of someone diagnosed with appendicitis', 'A clinician or hospital staff', and 'Other'. A 'reset' link is at the bottom right of this section. The second question is 'What emergency department or hospital did you go to for this diagnosis?' with a red asterisk and 'must provide value'. To the right is a text input field with a dropdown arrow. A 'reset' link is at the bottom right of this section. The third question is 'Did your hospital tests reveal an appendicolith, a stone-like object in your appendix? Your doctors can clarify if you're uncertain, or you can just select 'I Don't Know'.' with a red asterisk and 'must provide value'. To the right are three buttons: 'Yes', 'No', and 'I Don't Know'. A 'reset' link is at the bottom right of this section. At the bottom, a large light blue box contains the text: 'Next, learn about appendicitis and your treatment options (6-minute video)'.

TRIAD

English

Page 2 of

Are you a:
* must provide value

Patient diagnosed with appendicitis

Family member or loved one of someone diagnosed with appendicitis

A clinician or hospital staff

Other

reset

What emergency department or hospital did you go to for this diagnosis?
* must provide value

Did your hospital tests reveal an appendicolith, a stone-like object in your appendix? Your doctors can clarify if you're uncertain, or you can just select 'I Don't Know'.
* must provide value

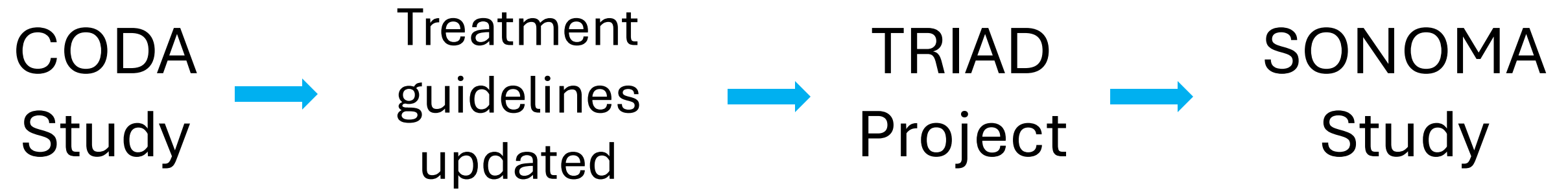
Yes

No

I Don't Know

reset

Next, learn about appendicitis and your treatment options (6-minute video)





Selection and Outcomes of Patients with Non-Operatively Managed Appendicitis (SONOMA)

A retrospective multicenter cohort study assessing the frequency of NOM, which patients are selecting NOM, and details of the outcomes and healthcare utilization of these patients.

- Focusing on patient selection, post-treatment outcomes, trends of NOM usage across TRIAD sites
 - Post-treatment outcomes gathered via Electronic Medical Records (EMRs)
-

Study population

- Adults with **first episode Acute Appendicitis**
 - Encounter dates: Jan 1, 2023 – Jul 31, 2024
 - (~9 months pre-TRIAD and ~9 months post-TRIAD)
 - ICD-10 **K35 / K36 / K37** for appendicitis without complication
 - Treatment pathways of **Appendectomy** or **antibiotics** as primary treatment
 - Evidence of antibiotics use if possible
- Sites
 - 13 hospitals including UW Harborview, UW Montlake, Michigan, etc.
 - All TRIAD participating hospitals

Pregnant and Immunosuppressed Individuals

- Treated as separate populations that will have additional details collected on their course if they have complicated appendicitis
- Representation of these populations is low in the study; therefore, they will be removed from our analysis
- More for exploratory purpose that takes advantage of a large cohort; A different study is planned soon to assess the outcomes of these specific populations

Missing Data

- Examine Missingness pattern
- Assess Feasibility of Complete Case Analysis
 - Key covariates have modest missingness ($\approx \leq 10\text{--}15\%$).
 - Missingness is plausibly MAR conditional on observed covariates.
 - Complete and incomplete cases do not show major clinical or demographic imbalance.
- Imputation (Sensitivity Analysis)
 - **MI** when important predictors/covariates show higher missingness or missingness correlates with observed variables.

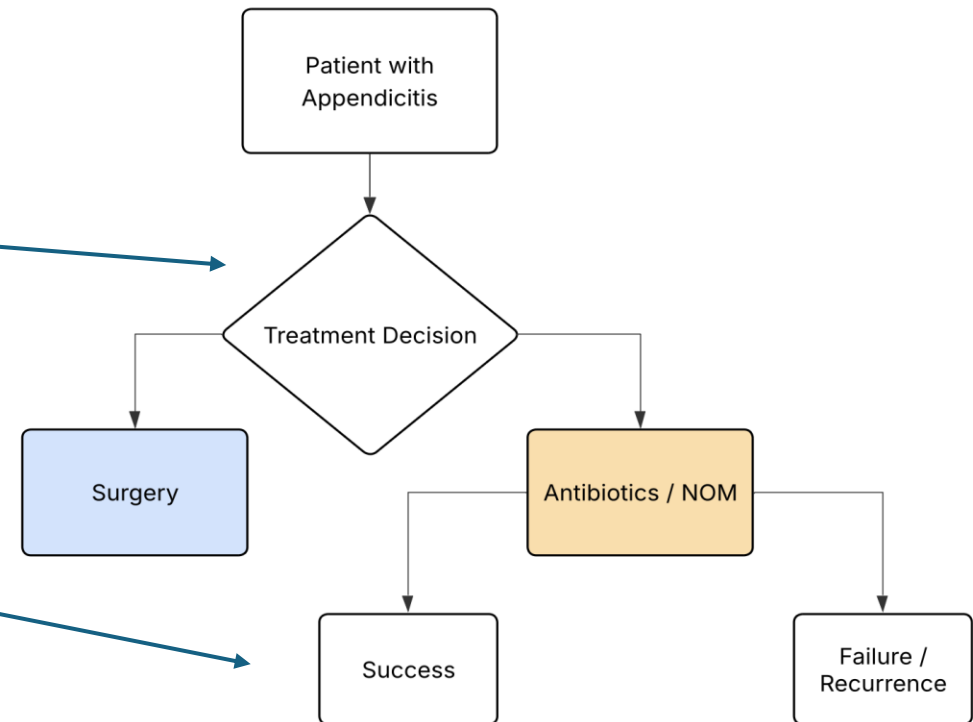
Our focus

Patient Selection

- What are patients choosing for treatment?

Patient Outcome

- What happens to patients after NOM?



Patient Selection Aims

Primary Study Aim

- Determine what socio-demographic, clinical and site factors impact treatment decision between surgery and NOM.

Secondary Study Aim

- Determine if the introduction of the TRIAD Decision Support Tool, AppyOrNot, changed the proportion of patients choosing NOM for each site.

S.Aim 1: Determine if socio-demographic, clinical, and site factors impact treatment decision

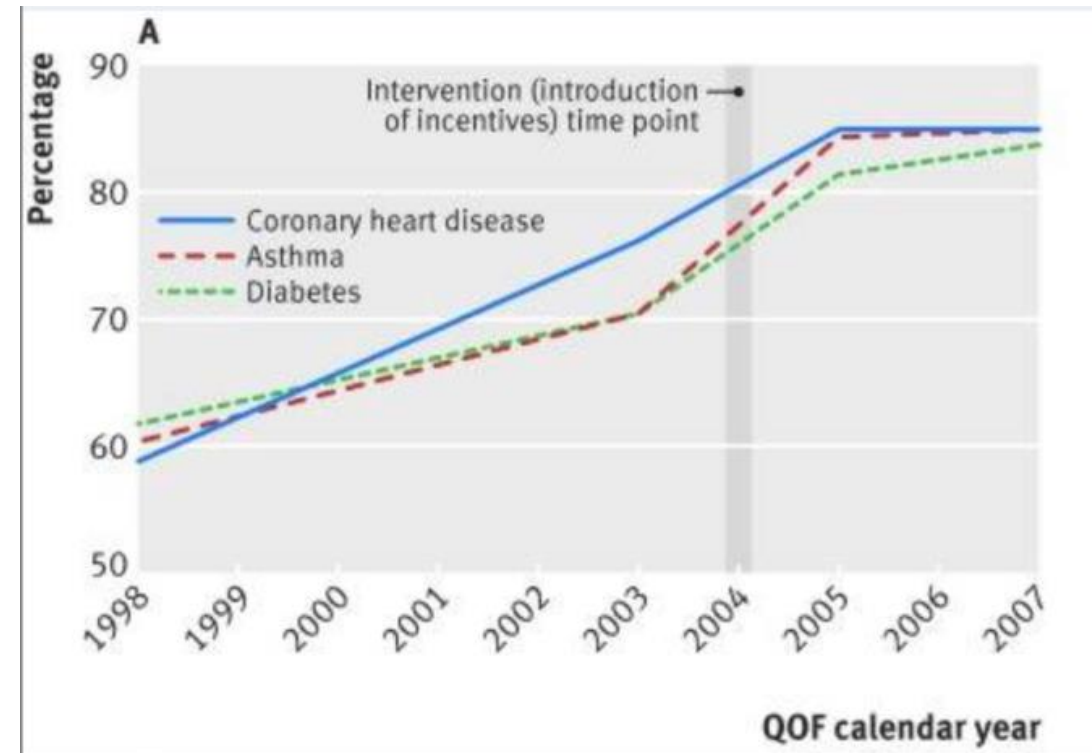
- **Outcome:** NOM vs. appendectomy selected at initial encounter
- **Model:** Logistic mixed-effects model (site as random effect)
- **Covariates:** Age, site, sex, appendicolith presence, NLR, comorbidity index, appendix diameter, race, ethnicity, language barrier indicator
- **Report:**
 - Odds ratios, CIs, and Wald test p-values (sig = 0.05) via forest plot
 - Proportions of NOM per site

S.Aim 2: Determine if the introduction of the Decision Support Tool, AppyOrNot, changes the proportion of patients choosing NOM for each site.

Outcome: Monthly proportion of patients receiving NOM each month.

Model: Interrupted Time Series (ITS) model to estimate monthly NOM proportions using Poisson (or Negative Binomial*) regression.

Covariates: time (in months), an indicator of the site's usage of AppyOrNot (pre/post), (usage*time) interaction term, site (random intercept), an offset of total appendicitis cases per month.

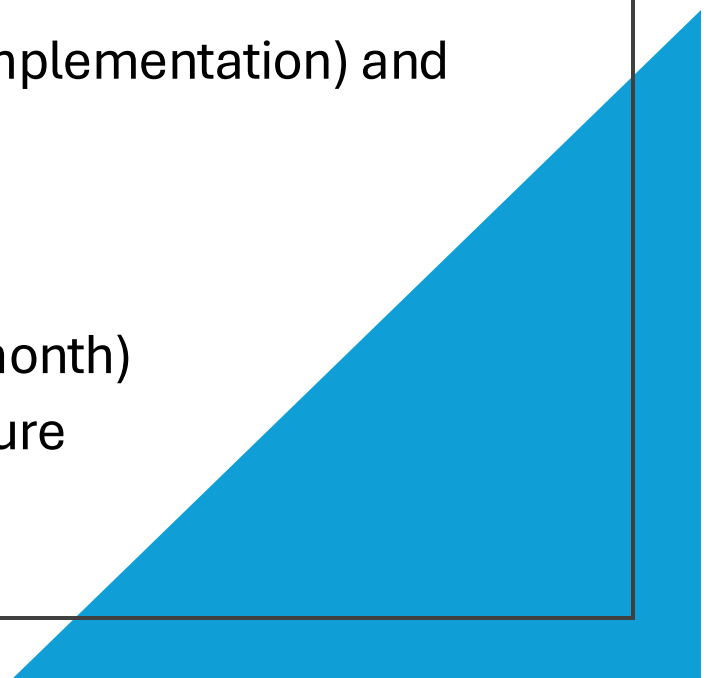


Pre/post implementation of AppyOrNot

The implementation start month for each site will be defined using the first month with sustained AppyOrNot use (based on site-level tool usage/click logs).

- If implementation timing is the same across sites:
Use a binary Post indicator(0=pre-implementation, 1=post-implementation) and $\text{Time} \times \text{Post}$ interaction to model level and slope change.
- If implementation timing differs across sites:
Create a relative time variable:
$$\text{EventTime} = (\text{current month}) - (\text{site's implementation month})$$

Model effect using relative time (linear or categorical) to capture dynamic changes around implementation.



Patient Outcome Aims

Primary Study Aim

- Determine what factors are associated with appendectomy after NOM, particularly early, non-elective appendectomy after NOM.

Secondary Study Aims

- Determine median time to first recurrence of appendicitis symptoms after NOM and median time to appendectomy after NOM.
- Investigate the role of inflammation on initial appendicitis and appendicitis after NOM

O.Aim 1: Determine what factors are associated with appendectomy after NOM

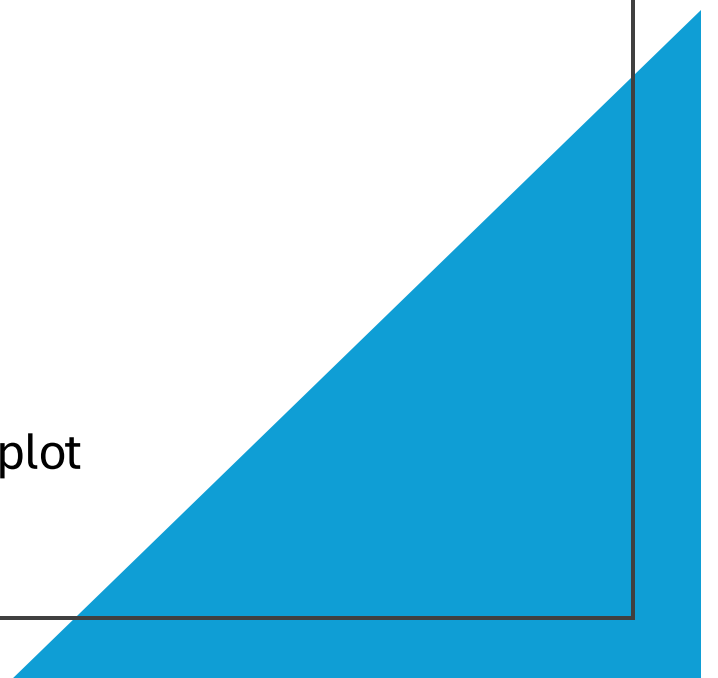
Outcome:

- no appendectomy after NOM (baseline)
- elective appendectomy after NOM
- urgent & late appendectomy after NOM
- urgent & early appendectomy after NOM

Model: Multinomial logistic regression with site as fixed effect

Covariates: Same as Aim 1, socio-demographics and clinical factors

Report: Odds ratios, CIs, and Wald test p-values (sig = 0.05) via forest plot



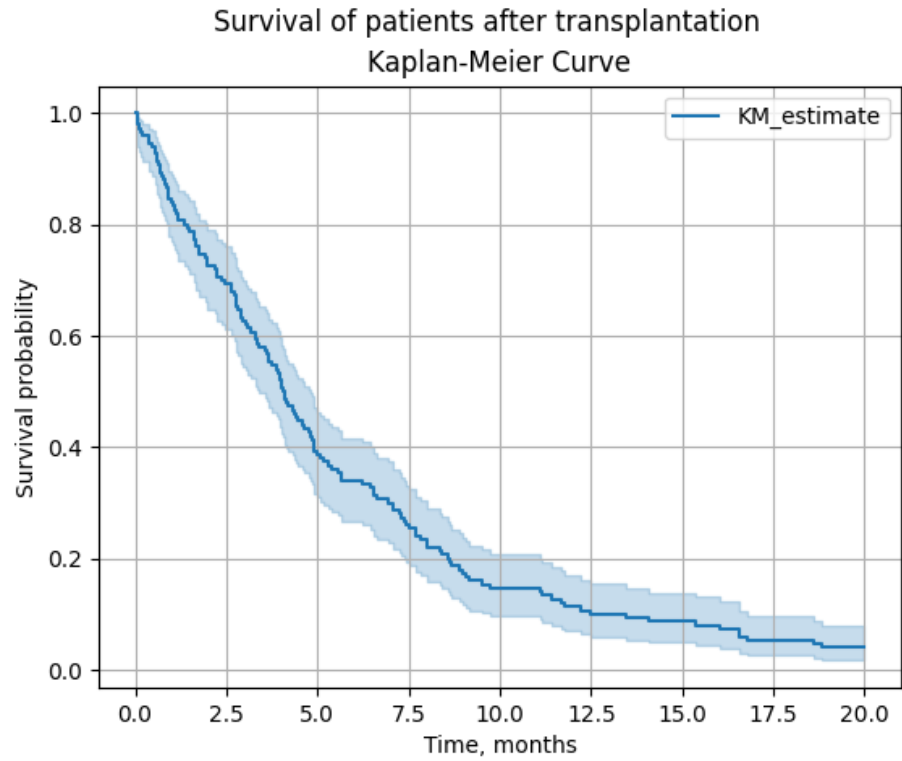
O.Aim 2: Determine median time to first recurrence of appendicitis symptoms after NOM and median time to appendectomy after NOM

Individuals who did not experience the event in the first year after NOM are administratively censored

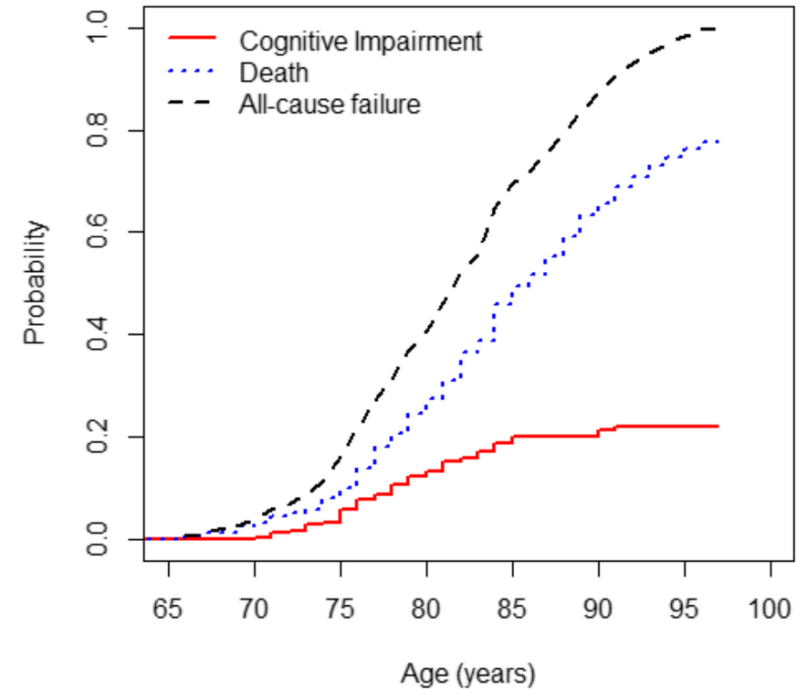
- Kaplan Meier estimator of probability of no recurrence of appendicitis symptoms after NOM
- Nelson-Aalen estimator of total risk of recurrence of appendicitis symptoms over time

May explore stratifying by site

- Aalen-Johansen estimator of competing risks of non-elective vs elective appendectomy after NOM over time



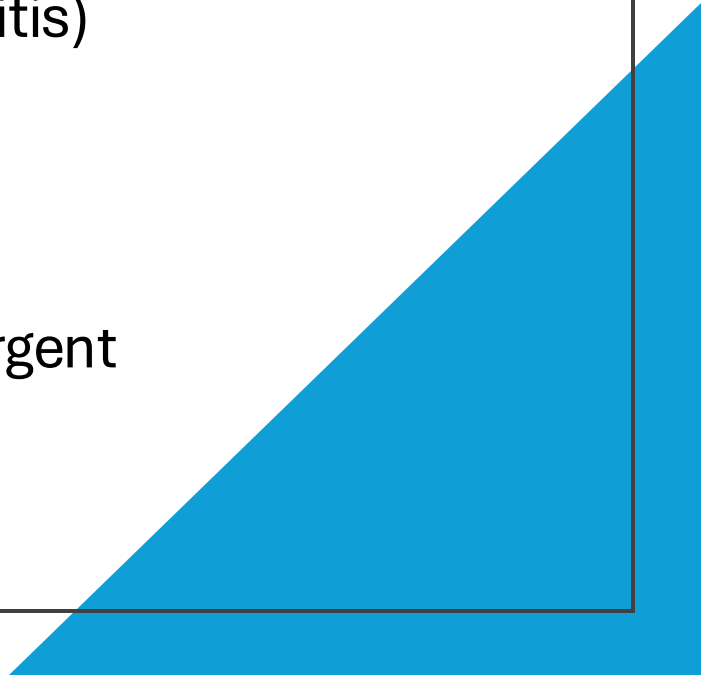
Kaplan Meier Estimator
of Survival Function



Aalen-Johansen Estimator of
Cumulative Incidence Function

O.Aim 3: Investigate the role of inflammation on initial appendicitis and appendicitis after NOM.

- 3a. Determine a NLR threshold that could indicate complicated appendicitis (vs non-complicated appendicitis) *at index*, compare it with past literature, and test its predictive ability.
- 3b. Test our inflammation cutoff for detecting early and urgent appendicitis *after NOM*



O.Aim 3a: Develop and test NLR threshold at index

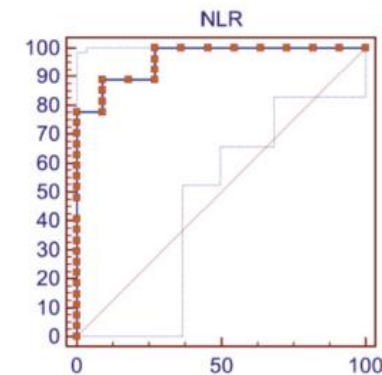
1. Using ROC analysis, compute optimal cutoff and 95% CI via Youden index.
2. Fit logistic regression with cutoff as predictor and complicated vs non-complicated as outcome; report OR and 95% CI to determine effectiveness of threshold.
3. Compare with previous literature

Dölling, M., et al. “Complicated Appendicitis: Value of Inflammatory Markers Based on EAES 2015 Guidelines.” *Surgical Endoscopy*, vol. 39, 2025, pp. 6586–6596.

ROC curve analysis

Appendicitis versus no appendicitis. ROC curve analysis identified NLR of 4.7 as cut-off value for diagnosing appendicitis with sensitivity of 88.89% (95% CI 70.8%–97.6%) and specificity of 90.91% (95% CI, 58.7%–99.8%). AUC was 0.96 (95% CI 0.84–1.0, $P < 0.0001$) (Fig. 4A).

A) Appendicitis versus no appendicitis



O.Aim 3b: Test new inflammation cutoff for detecting early and urgent appendicitis *after NOM*

- **Model:** Logistic Regression
 - **Outcome:** early/urgent appendectomy after NOM vs all other outcomes
 - **Covariate:** NLR cutoff from 4.a
 - **Report:** Odds ratios, CIs, and Wald test p-values (sig = 0.05) for high NLR
 - Could have use for doctors in determining NOM vs appendectomy.
- 
- A large blue right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

Sites expected to finalize data abstraction by Jan 1st

- University of Washington
- University of Michigan
- Columbia University

Sites expected to finalize data abstraction after Jan 1st

- University of Tennessee
- Northwestern University
- Grady Health (Atlanta, GA)

Given the unknown timing of abstraction finalization for various sites in the study, our analysis will likely be an interim analysis on the sites abstracted by the start of the winter quarter



Thank you!

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